



ReasonAudit: Evaluating Error Awareness and Self-Correction in Large Language Models

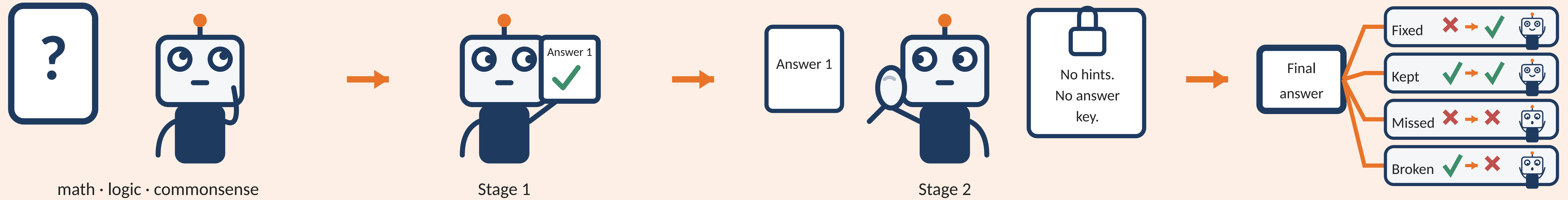
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Can an LLM catch its own mistakes, or does second-guessing make things worse?



We measure how often self-review fixes errors, and how often it ruins correct answers.

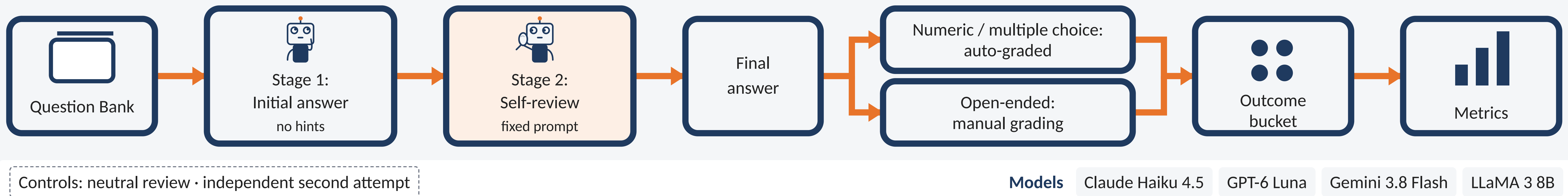
Motivation

- LLMs are used in coding, teaching, and research, often with little human checking.
- They can give wrong answers with full confidence.
- Self-review is fragile: models may miss errors, keep bad answers, or change correct answers to wrong ones.
- We need a precise measure of when self-correction helps and when it hurts.

Objectives

- Build a labeled reasoning dataset with verified answers and difficulty tags.
- Measure Correction Rate, Damage Rate, and accuracy change after self-review.
- Compare open-weight and proprietary models across reasoning types.
- Link difficulty factors to self-correction success.

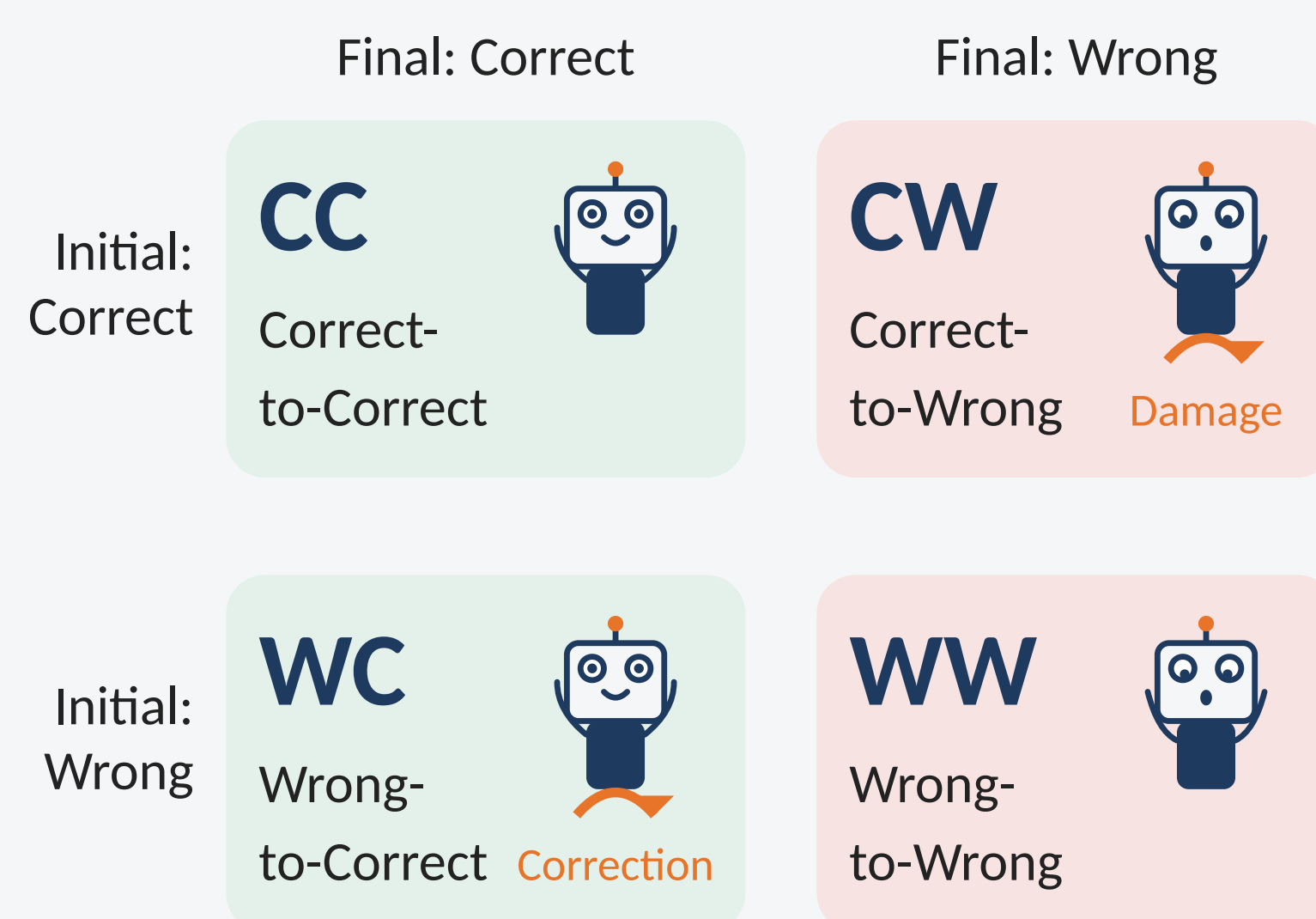
Methodology



Dataset

- Domains: Mathematics, Formal Logic, Commonsense
- Difficulty modifiers: each question tagged yes / no
- Ambiguity, Missing information, Contradiction, Distracting information
- Domains: Mathematics · Formal Logic · Commonsense
 - Difficulty modifiers (yes/no tag per question): Ambiguity · Missing information · Contradiction · Distracting information
 - Built by: collecting questions from public datasets, then manual annotation and tagging. Each question keeps its source and a flag if modified.
 - Scale: 20-question pilot, designed to scale to 2,000.

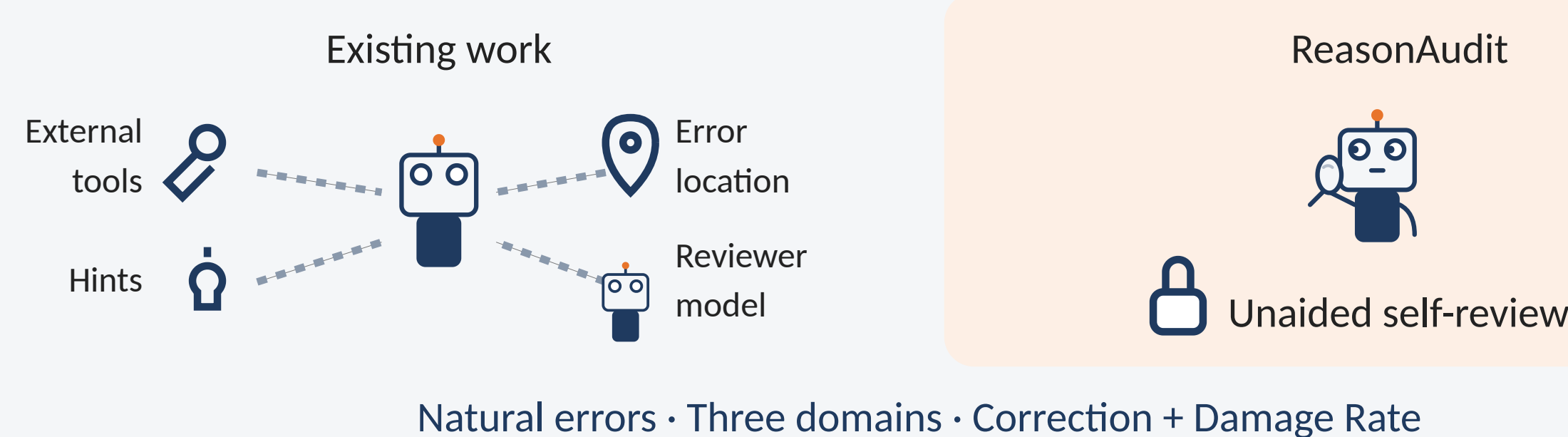
Outcome Buckets



Metrics

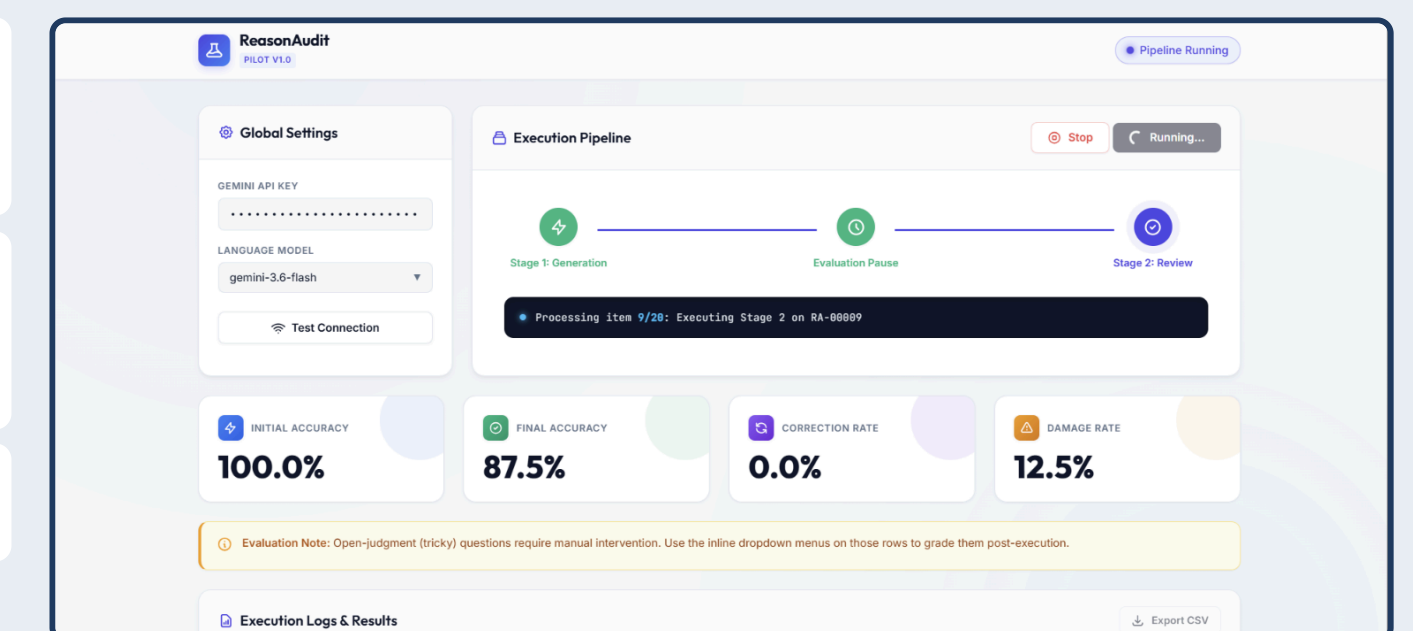
- Correction Rate** = $WC / (WC + WW)$: share of wrong answers the model fixed
- Damage Rate** = $CW / (CC + CW)$: share of correct answers the model ruined
- Accuracy Change** = Final Accuracy - Initial Accuracy
- Analysis:** McNemar's test, 95% confidence intervals, breakdown by domain and difficulty tag.

Research Gap

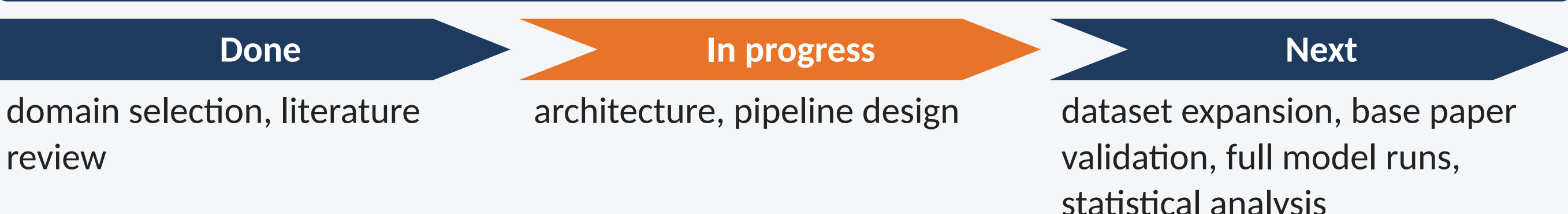


Pilot System

- Two-stage pipeline in Google Sheets + Apps Script, with a web pilot tool.
- Logs every prompt and raw response for traceability.
- Automatic retry on rate limits, CSV export.



Status and Next Steps



References

- Huang et al., Large Language Models Cannot Self-Correct Reasoning Yet, ICLR 2024
- Tyen et al., LLMs Cannot Find Reasoning Errors, but Can Correct Them Given the Error Location, ACL Findings 2024
- Tsui, Self-Correction Bench, COLM 2026
- Madaan et al., Self-Refine, NeurIPS 2023
- Yuan et al., Hidden Error Awareness in Chain-of-Thought Reasoning, 2026